ENGR 1451/2451: Exploratory Data Science

A Closer Look at Tests of Significance

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Today: What Does a P-Value Really Tell Us?

How to interpret the results—and when the tests are misleading.

- Explain why the 5% cutoff is not a sharp boundary.
- Recognize **p-hacking** (data snooping) and the danger of many tests.
- Distinguish **statistical significance** from **practical importance**.
- Check whether a **chance model** and study design justify a P-value.

Does $P = 0.049$ Differ From $P = 0.051$?

Study A

$$P = 0.049$$

Called **significant** at the 5% level.

Study B

$$P = 0.051$$

Called **not significant** at the 5% level.

Should these studies lead to very different scientific conclusions?

P-Value

Assume the null hypothesis and chance model are correct

$$P = \Pr(\text{result at least as extreme as observed} \mid H_0)$$

It answers

How unusual are these data **under the assumed null model**?

It does NOT answer

What is the probability H_0 is true? Is the effect important? What caused it?

A 5% Test Has False Alarms

Suppose all 100 null hypotheses are true

At the 5% significance level, the **expected number of false positives** is

$$100 \times 0.05 = \boxed{5}.$$

It may be 2, 5, 8, or another number in a particular set of tests.

100 tests; 5 colored as an *illustration*.

"Statistically significant" does not mean "impossible by chance."

A Rare Cancer Cluster Isn't Always Surprising

A town of 10,000 people

Suppose the chance of **two or more liver-cancer cases** in a particular town in one year is about **0.5%**, under the baseline chance model.

One town in one year

$$P \approx 0.005$$

Unusual *if selected beforehand*.

100 towns over 10 years

$$100 \times 10 = 1{,}000 \text{ town-years}$$

$$1{,}000 \times 0.005 = \boxed{5}$$

expected chance clusters

Key idea: A cluster found *after searching everywhere* is less surprising than one predicted for a specific town and year.

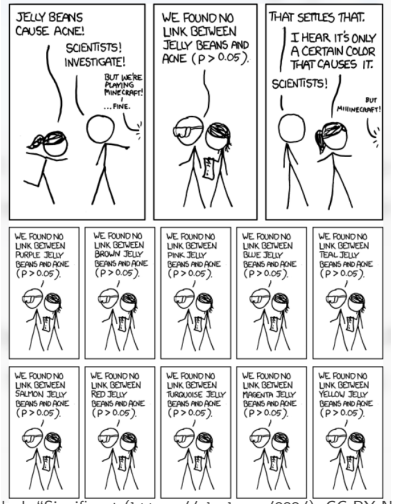
P-Hacking (Data Snooping)

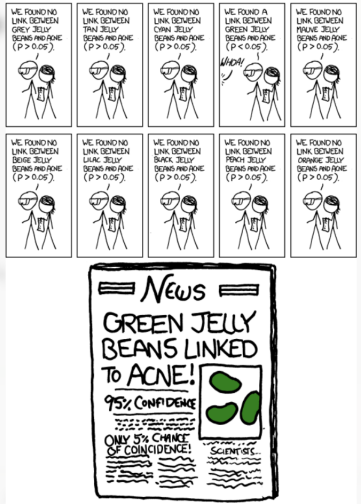
Chemical screening experiment

Investigators compare cancer rates in roughly **25 organs and organ systems** in laboratory mice. The liver gives $z \approx 2.4$, with **one-tailed** $P \approx 1\%$.

- A P-value for **one selected organ** ignores the other searches.
- The chance of *some* striking result increases when many outcomes are examined.
- Testing a new sample can check whether the finding replicates.

P-Hacking





Source: xkcd, "Significant (`https://xkcd.com/882/`); CC BY-NC 2.5.

Bonferroni Correction: Multiple Tests

Testing many hypotheses increases the chance of finding at least one false positive.

One test

$$\alpha = 0.05 = 5\%$$

A result with $P < 0.05$ is significant.

Bonferroni: K tests

$$\boxed{\alpha_{\text{each}} = \frac{\alpha}{K}}$$

For $K = 25$: $\alpha_{\text{each}} = 0.002$.

Example: With 25 organ tests, the liver's $P \approx 1\%$ is below 5% but **not** below 0.2%.

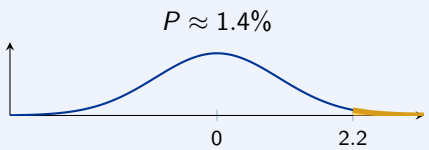
Why adjust? Bonferroni limits the probability of at least one false positive across the K tests to at most α, assuming valid individual tests.

Discuss: If only the liver outcome had been planned and tested, would a multiple-testing adjustment be needed?

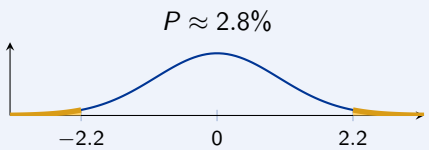
One-Tailed Versus Two-Tailed: The Same 61 Heads

Coin tossed 100 times: 61 heads. Under $H_0 : p = 0.50$, expected 50, $\mathrm{SE} = 5$, and $z = (61 - 50)/5 = 2.2$.

One-tailed: $H_A : p > 0.50$



Two-tailed: $H_A : p \neq 0.50$



Same Drug Trial, Two Different P-Values

Cholestyramine trial

Randomized, double-blind trial; heart attacks over 7 years:

	Treated	Control
Heart attacks	155/1,906	187/1,900
Rate	8.1%	9.8%

If planned: decrease

$H_A: \ p_T < p_C$ (one-tailed) $\qquad P \approx 3.5\%$

Any difference?

$H_A: \ p_T \neq p_C$ (two-tailed) $\qquad P \approx 7\%$

The alternative hypothesis including its direction must be chosen before inspecting the results.

Statistical Significance Is Not Practical Importance

Statistical question

Is the observed difference **hard to explain by chance variation**, under the assumed model?

$$\frac{\text{difference}}{\text{SE}}$$

Practical question

Is the difference **large enough to matter** for the real-world decision?

Consider units, cost, risk, and context.

A tiny effect can be statistically significant if it is measured precisely enough.

A Tiny P-Value for a Tiny Difference

Independent random samples of 2,500 city children and 2,500 rural children take a 40-word vocabulary test.

Observed scores

	City	Rural
Mean	26	25
SD	10	10
Sample size	2,500	2,500

Test of equal population means

$$SE_{\text{diff}} = \sqrt{\frac{10^2}{2500} + \frac{10^2}{2500}} \approx 0.283$$

$$z = \frac{26 - 25}{0.283} \approx 3.54$$

$$\boxed{P \approx 0.04\% \text{ (two-sided)}}$$

Two different conclusions:

- **Statistical significance:** Very strong evidence that the population means differ ($P < 0.05$).
- **Practical significance:** The observed difference is only **1 point on a 40-word test** (maximum: 80 points).

Discuss: Does this tiny P-value justify a major educational intervention to close the gap?

The Same Difference Can Become Significant

Independent random samples have averages of 49 and 51, with SD 10 in each group.

	$n = 100$ **per group**	$n = 400$ **per group**
Observed difference	2 points	2 points
SE of difference	1.41	0.71
$\lvert z \rvert$	1.41	2.83
Two-tailed P	≈ 0.16	≈ 0.0047
Verdict at 5%	Not significant	Significant

Larger samples reduce uncertainty about the difference; they do not increase the observed difference itself.

Discussion: Can Any Small Difference Become Significant?

Question

If an estimated difference stays at **two points**, why can raising the sample size change the conclusion of a significance test?

Discuss: What would you want to see in a scientific paper *besides* the P-value?

Not Significant Does Not Mean No Effect

An election study

A statistical model estimates inflation's effect on candidate vote share as **+7 percentage points**, with an **SE of 5 points**.

One possible test

$$z = 7/5 = 1.4$$

Two-tailed $P \approx 0.16$

Approximate 95% interval

$$7 \pm 2(5)$$

$[-3, 17]$ points

Discuss: The researchers conclude "inflation had no effect." Is that justified?

Before Calculating P: What Is Random?



- **Selection:** Were observations randomly sampled or treatments randomly assigned?
- **Dependence:** Are repeated measurements or groups being treated as independent?
- **Assumptions:** Does the model justify the standard error and reference distribution?

Example: Should We Test the Census?

U.S. population aged 65 or older

	1980	2000
Population	227 million	281 million
Age 65 or older	11.3%	12.3%

The proportion increased by **1 percentage point**. Are these data just a random sample of the two populations?

Discuss: Would a two-sample significance test add meaningful information?

P-Values and Nonrandom Applicants

Pitt first-year admissions, Fall 2025

Pittsburgh campus:

Men: **56.4%** admitted ($n = 27{,}732$)
Women: **61.7%** admitted ($n = 36{,}992$)

Discuss: What further information would be needed to explain why the rates differ?

Source: University of Pittsburgh Common Data Set 2025–26, Sec. C1.

Are 112 Comparisons Really 112 Independent Trials?

Employee salaries

A company employs **7 men and 16 women**. Of the $7 \times 16 = 112$ male–female pairs, the man earns more in **68 pairs**.

A tempting calculation

Assume each pair is an independent 50–50 trial:

$$E = 56, \qquad \mathrm{SE} \approx 5.3$$

$$z \approx \frac{68 - 56}{5.3} \approx 2.3$$

What is being counted?

The calculation makes **112 pairwise comparisons** from **23 employees**. What assumptions are needed for the reported standard error?

Discuss: Is the calculated *one-sided* $P \approx 1.2\%$ valid evidence of discrimination? Why or why not?

Does $P = 0.02\%$ Prove ESP?

The "Aquarius" experiment

A machine selects one of four targets; participants guess **2,006 of 7,500** correctly, versus **1,875** expected under independent, equally likely targets.

$$z \approx 3.5, \qquad \text{one-tailed } P \approx 0.02\%.$$

Discuss: What else should investigators check before attributing the result to ESP?

A tiny P-value does not validate the apparatus or identify the cause.

Summary and Overview

What a significance test does

It asks whether a difference is unusual **under a specified null hypothesis and chance model**—not whether the difference matters or what caused it.

Interpret the result

- The **5% cutoff** is a convention; a significant result can still be chance. Report actual P.
- **P-hacking** (Data snooping) and multiple tests make small P-values easier to find; choose the tail in advance.
- **Sample size changes** P. A small P means statistical significance, not necessarily practical significance.

Check what the test assumes

- A credible **chance model** for the data is essential; beware census and convenience data.
- A test does **not validate design or prove cause**; a large P does not prove no difference.
- Prefer **difference estimates, SEs/intervals, and practical interpretation** alongside P.